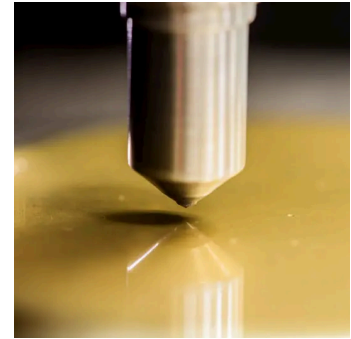


Hardness Testing

Understanding hardness:

We all intuitively understand hardness. Cheese is easier to dent than steel. Definitionally, hardness is the measure of resistance to localized plastic deformation.

Hardness is generally related to the *strength of intermolecular bonds* (think strength of carbon-carbon bonds in diamond, or strength of ionic bonds in ceramics), but differs based on many material properties.

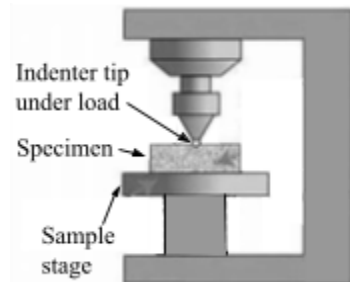


There are three main types of hardness measurements:

Scratch hardness	Indentation hardness	Rebound hardness
A measure of a sample's resistance to deformation due to friction from a hard object Ex: Mohs scale	A measure of a sample's resistance to deformation due to a compressive load from a sharp object Ex: Rockwell, Vickers	A measure of the height of the bounce of a diamond-tipped hammer dropped onto a material Ex: Leeb rebound hardness

The Breakerspace hardness tester tests indentation hardness, using the following methods.

In each method, an indenter tip is pressed into your sample with a heavy load. The resulting dent is measured. Then, a hardness value is calculated.



Procedure	What does it measure?	What makes it unique?
Vickers	Surface area of indentation	Has a square pyramidal tip, not a sphere. Vickers tests with different loads result in the same hardness values
Rockwell	Depth of the dent	Has subscales: HRA through HRK, each using different indentation tips and loads for different materials. A conversion table is needed to convert between scales.
Brinnell	Diameter of the dent	Similar applications to Rockwell, but covers the whole range of hardness on one scale.

Instructions on using the Breakerspace Hardness Tester

Sample Preparation:

- Prepare a flat, ideally polished sample.

Instrument Start-up

- Turn on machine with red switch in back.
- Log in to the Impressions software. The username is “mit hardness” and leave the password field empty (no password)

Instrument Operation

- Position your sample on the sample stage
- Find a clean spot on your sample using the camera. Use the joystick or the software to move the sample stage.
- Get into focus with 2.5x, then change 10x and refocus.
 - Some hardness testing procedures, like Vickers need to be zoomed in, because they use the camera to determine the boundaries of each indent.
- Click In Focus button
- Choose your hardness test using the top left blue box.
 - When it comes to tests, which offer different loads (i.e. Vickers 1kgf versus 5kgf), start low and work your way up
- Click Start and the testing procedure will proceed automatically
- To save your data, (one method) click Report on the bottom left, then Print, then Export, then to pdf, name your file, then OK. Your file should pop-up in C:\Users\HardnessTester\Desktop.



Considerations!

- To make a pattern, click “Pattern” and choose your settings. Make sure to return to single point when you’re done using the pattern!
- Make sure your tip matches your surface. A too soft tip for your surface may damage the machine. For example, HRA uses a harder tip than HRB.

Instrument shut-down

- Remove your sample from the machine and turn off the workstation.
- Toggle the power switch in back of the machine to off

See the Breakerspace website for the extended SOP! Each workstation also has a desktop icon linking to its SOP.

As always, if you need help, contact a lab assistant!